SQL Assessment

Q1. Daily Agent Productivity

*1st 20 rows.

```
SELECT
    d.work_date,
    d.agent_id,
    d.agent_name,
    h.hub_name,
    d.assigned,
    d.delivered,
    d.holds,
    d.returned

FROM daily_productivity d

JOIN hubs h
    ON d.hub_id = h.hub_id

ORDER BY d.work_date, d.agent_id

LIMIT 20;
```

work_date	agent_id	agent_name	hub_name	assigned	delivered	holds	returned
2026-10-13	A1001	Weno Tremaine	Greenfield	20	18	2	0
2026-10-13	A1002	Valel Wexley	Greenfield	21	17	4	0
2026-10-13	A1003	Torian Vantell	Greenfield	27	22	3	2
2026-10-13	A1004	Garian Larkin	Greenfield	20	19	1	0
2026-10-13	A1005	Wenan Brannick	Greenfield	23	20	2	1
2026-10-13	A1006	Toren Rowntree	Greenfield	26	23	2	1
2026-10-13	A1007	Coris Rowntree	Greenfield	27	25	0	2
2026-10-13	A1008	Garon Holloway	Greenfield	26	22	3	1
2026-10-13	A1009	Perian Halvorsen	Greenfield	34	27	6	1
2026-10-13	A1010	Joris Fenwick	Greenfield	28	23	2	3
2026-10-13	A1011	Marel Fenwick	Greenfield	14	12	1	1
2026-10-13	A1012	Zarik Rowntree	Lakeview	19	18	1	0
2026-10-13	A1013	Pero Ketteridge	Lakeview	25	22	2	1
2026-10-13	A1014	Zaris Quorne	Lakeview	26	23	2	1
2026-10-13	A1015	Toren Larkin	Lakeview	24	21	3	0
2026-10-13	A1016	Noren Vantell	Lakeview	17	15	0	2
2026-10-13	A1017	Joran Pelham	Lakeview	23	23	0	0
2026-10-13	A1018	Lorus Brannick	Lakeview	14	13	1	0
2026-10-13	A1019	Selus Rowntree	Lakeview	19	17	2	0
2026-10-13	A1020	Noris Fenwick	Lakeview	15	14	1	0

*Average deliveries

```
SELECT
    h.hub_name,

    SUM(d.delivered) AS total_deliveries,

    COUNT(*) AS available_agent_days,

    ROUND(
        SUM(d.delivered) / COUNT(*),
        2
    ) AS average_deliveries

FROM daily_productivity d

JOIN hubs h
    ON d.hub_id = h.hub_id

GROUP BY h.hub_id, h.hub_name

ORDER BY h.hub_id;
```

	hub_name	total_deliveri...	available_agent_da...	average_deliveries	
	Greenfield	3908	147	26.59	
	Lakeview	3001	154	19.49	
	Riverside	3479	168	20.71	
	Hillcrest	3798	126	30.14	
	Brookside	3411	154	22.15	
	Westwood	3481	133	26.17	

Q2. Hub Utilisation in Week 1 and Week 2

```
SELECT
    h.hub_name,
    d.week_no,

    COUNT(DISTINCT d.agent_id) AS available_agents,
```

```
    SUM(d.assigned) AS total_attempts,

    ROUND(
        SUM(d.assigned) / (COUNT(*) * 40) * 100,
        2
    ) AS utilisation_percentage,

    CASE
        WHEN d.week_no = 2
         AND SUM(d.assigned) / (COUNT(*) * 40) > 0.85
        THEN 'Above target'
        ELSE 'Within target'
    END AS target_status

FROM daily_productivity d

JOIN hubs h
    ON d.hub_id = h.hub_id

GROUP BY h.hub_id, h.hub_name, d.week_no

ORDER BY h.hub_id, d.week_no;
```

hub_name	week_no	available_agents	total_attem...	utilisation_percenta...	target_stat...
Greenfield	1	11	1925	62.50	Within target
Greenfield	2	10	2482	88.64	Above target
Lakeview	1	11	1647	53.47	Within target
Lakeview	2	11	1730	56.17	Within target
Riverside	1	12	2281	67.89	Within target
Riverside	2	12	3051	90.80	Above target
Hillcrest	1	9	1943	77.10	Within target
Hillcrest	2	9	2416	95.87	Above target
Brookside	1	11	2023	65.68	Within target
Brookside	2	11	2202	71.49	Within target
Westwood	1	10	1870	66.79	Within target
Westwood	2	9	2025	80.36	Within target

Utilisation = attempts ÷ (available agent-days × 40)

Q3. Hold Reasons

- Total holds and hold and agent side percentage .

```
SELECT
   h.hub_name,

   COUNT(*) AS total_holds,

   SUM(
      CASE
         WHEN l.hold_reason IN (
            'Late Dispatch',
            'Not Attempted',
            'Agent Time Shortage'
         )
         THEN 1
         ELSE 0
      END
   ) AS agent_side_holds,

   ROUND(
      SUM(
         CASE
            WHEN l.hold_reason IN (
               'Late Dispatch',
               'Not Attempted',
               'Agent Time Shortage'
            )
            THEN 1
            ELSE 0
         END
      ) / COUNT(*) * 100,
      2
   ) AS agent_side_percentage

FROM order_logs l

JOIN orders o
   ON l.order_id = o.order_id

JOIN hubs h
   ON o.hub_id = h.hub_id

WHERE l.status = 'Hold'
```

```
  AND l.event_time >= '2026-10-20'
  AND l.event_time < '2026-10-27'

GROUP BY h.hub_id, h.hub_name

ORDER BY h.hub_id;
```

	hub_name	total_holds	agent_side_hol...	agent_side_percenta...	
	Greenfield	168	32	19.05	
	Lakeview	123	26	21.14	
	Riverside	874	563	64.42	
	Hillcrest	182	40	21.98	
	Brookside	186	49	26.34	
	Westwood	146	30	20.55	

- Top 2 hold reasons

```
WITH reason_counts AS (
    SELECT
        h.hub_id,
        h.hub_name,
        l.hold_reason,
        COUNT(*) AS total_holds

    FROM order_logs l

    JOIN orders o
        ON l.order_id = o.order_id

    JOIN hubs h
        ON o.hub_id = h.hub_id

    WHERE l.status = 'Hold'
      AND l.event_time >= '2026-10-20'
      AND l.event_time < '2026-10-27'

    GROUP BY h.hub_id, h.hub_name, l.hold_reason
),

ranked_reasons AS (
    SELECT
        *,
```

```
        ROW_NUMBER() OVER (
            PARTITION BY hub_id
            ORDER BY total_holds DESC, hold_reason
        ) AS reason_rank

    FROM reason_counts
)

SELECT
    hub_name,
    hold_reason,
    total_holds

FROM ranked_reasons

WHERE reason_rank <= 2

ORDER BY hub_id, reason_rank;
```

hub_name	hold_reason	total_holds
Greenfield	Customer Unreachable	73
Greenfield	Customer Rescheduled	35
Lakeview	Customer Unreachable	50
Lakeview	Customer Rescheduled	25
Riverside	Late Dispatch	312
Riverside	Customer Unreachable	156
Hillcrest	Customer Unreachable	77
Hillcrest	Customer Rescheduled	39
Brookside	Customer Unreachable	68
Brookside	Customer Rescheduled	43
Westwood	Customer Unreachable	66
Westwood	Customer Rescheduled	28

Q4. What Happens to Held Orders

```
SELECT
   h.hub_name,

   COUNT(*) AS held_orders,

   ROUND(
      SUM(CASE WHEN o.final_status = 'Delivered' THEN 1 ELSE 0 END)
      / COUNT(*) * 100,
      2
   ) AS delivered_percentage,

   ROUND(
      SUM(CASE WHEN o.final_status = 'Returned' THEN 1 ELSE 0 END)
      / COUNT(*) * 100,
      2
   ) AS returned_percentage,

   ROUND(
      SUM(CASE WHEN o.final_status = 'Hold' THEN 1 ELSE 0 END)
      / COUNT(*) * 100,
      2
   ) AS still_hold_percentage

FROM orders o

JOIN hubs h
   ON o.hub_id = h.hub_id

JOIN (
   SELECT DISTINCT order_id

   FROM order_logs

   WHERE status = 'Hold'
     AND event_time >= '2026-10-13'
     AND event_time < '2026-10-27'
) x

   ON o.order_id = x.order_id

GROUP BY h.hub_id, h.hub_name

ORDER BY h.hub_id;
```

hub_name	held_orders	delivered_percenta...	returned_percenta...	still_hold_percent...
Greenfield	303	88.78	2.64	8.58
Lakeview	228	89.47	4.39	6.14
Riverside	1184	76.27	13.94	9.80
Hillcrest	316	86.08	4.75	9.18
Brookside	324	80.25	11.42	8.33
Westwood	244	86.07	4.10	9.84

Q5. High-Return Agents

```
SELECT
    a.agent_id,
    a.agent_name,
    h.hub_name,

    DATEDIFF('2026-10-26', a.joining_date) AS tenure_days,

    SUM(d.assigned) AS total_attempts,

    SUM(d.returned) AS total_returns,

    ROUND(
        SUM(d.returned) / NULLIF(SUM(d.assigned), 0) * 100,
        2
    ) AS return_percentage

FROM daily_productivity d

JOIN agents a
    ON d.agent_id = a.agent_id

JOIN hubs h
    ON a.hub_id = h.hub_id

WHERE DATEDIFF('2026-10-26', a.joining_date) > 30

GROUP BY
    a.agent_id,
    a.agent_name,
    h.hub_id,
    h.hub_name,
```

```
    a.joining_date

HAVING SUM(d.returned) / NULLIF(SUM(d.assigned), 0) > 0.15

ORDER BY return_percentage DESC;
```

agent_id	agent_name	hub_name	tenure_days	total_attem...	total_retur...	return_percenta...
A1045	Belis Rowntree	Brookside	463	349	94	26.93
A1046	Coris Halvorsen	Brookside	647	414	91	21.98
A1047	Wenian Vantell	Brookside	526	413	72	17.43

Q6. Cost per Delivered Order

```
SELECT
    h.hub_name,
    d.week_no,

    SUM(d.delivered) AS total_delivered,

    SUM(
        CASE
            WHEN d.salary_type = 'Fixed'
            THEN 650
            ELSE 300 + d.delivered * 14
        END
    ) AS salary_cost,

    SUM(d.holds) * 30 AS hold_cost,

    SUM(d.returned) * 55 AS return_cost,

    ROUND(
        (
            SUM(
                CASE
                    WHEN d.salary_type = 'Fixed'
                    THEN 650
                    ELSE 300 + d.delivered * 14
                END
            )
            + SUM(d.holds) * 30
            + SUM(d.returned) * 55
```

```
        ) / NULLIF(SUM(d.delivered), 0),
        2
    ) AS cost_per_delivery

FROM daily_productivity d

JOIN hubs h
    ON d.hub_id = h.hub_id

GROUP BY h.hub_id, h.hub_name, d.week_no

ORDER BY h.hub_id, d.week_no;
```

hub_name	week_no	total_deliver...	salary_cost	hold_cost	return_cost	cost_per_deliv...
Greenfield	1	1698	48608	4770	3740	33.64
Greenfield	2	2210	48244	5040	5720	26.70
Lakeview	1	1469	48328	3630	3135	37.50
Lakeview	2	1532	48888	3690	4125	37.01
Riverside	1	1498	51338	18810	8580	52.56
Riverside	2	1981	54782	26220	10780	46.33
Hillcrest	1	1687	42196	4830	5225	30.97
Hillcrest	2	2111	47026	5460	6765	28.07
Brookside	1	1631	48286	5130	12155	40.20
Brookside	2	1780	48846	5580	12980	37.87
Westwood	1	1687	45192	3720	3245	30.92
Westwood	2	1794	42112	4380	4675	28.52

Q7. Agents Needed for the Forecast

```
WITH week2_workload AS (
    SELECT
        hub_id,
        SUM(assigned) AS attempts,
        COUNT(DISTINCT agent_id) AS available_agents

    FROM daily_productivity

    WHERE week_no = 2
```

```
    GROUP BY hub_id
),

week2_orders AS (
    SELECT
        hub_id,
        COUNT(*) AS new_orders

    FROM orders

    WHERE created_at >= '2026-10-20'
      AND created_at < '2026-10-27'

    GROUP BY hub_id
),

staffing AS (
    SELECT
        h.hub_id,
        h.hub_name,
        f.week_start,
        f.expected_orders,
        w.available_agents,

        CEIL(
            f.expected_orders
            * (w.attempts / n.new_orders)
            / 238
        ) AS required_agents

    FROM volume_forecast f

    JOIN hubs h
        ON f.hub_id = h.hub_id

    JOIN week2_workload w
        ON f.hub_id = w.hub_id

    JOIN week2_orders n
        ON f.hub_id = n.hub_id
)

SELECT
    hub_name,
```

```
    week_start,
    expected_orders,
    available_agents,
    required_agents,

    required_agents - available_agents AS staffing_gap

FROM staffing

ORDER BY hub_id, week_start;
```

hub_name	week_start	expected_ord...	available_agents	required_agen...	staffing_g...
Greenfield	2026-10-27	2470	10	12	2
Greenfield	2026-11-03	2545	10	12	2
Greenfield	2026-11-10	2570	10	12	2
Greenfield	2026-11-17	2495	10	12	2
Lakeview	2026-10-27	1544	11	7	-4
Lakeview	2026-11-03	1591	11	8	-3
Lakeview	2026-11-10	1606	11	8	-3
Lakeview	2026-11-17	1559	11	8	-3
Riverside	2026-10-27	2170	12	13	1
Riverside	2026-11-03	2230	12	14	2
Riverside	2026-11-10	2245	12	14	2
Riverside	2026-11-17	2200	12	13	1
Hillcrest	2026-10-27	2264	9	11	2
Hillcrest	2026-11-03	2333	9	11	2
Hillcrest	2026-11-10	2356	9	11	2
Hillcrest	2026-11-17	2287	9	11	2
Brookside	2026-10-27	1955	11	9	-2
Brookside	2026-11-03	2015	11	10	-1
Brookside	2026-11-10	2035	11	10	-1
Brookside	2026-11-17	1975	11	10	-1
Westwood	2026-10-27	1852	9	9	0
Westwood	2026-11-03	1909	9	9	0
Westwood	2026-11-10	1928	9	9	0
Westwood	2026-11-17	1871	9	9	0

Q8. The Hiring Request

I wouldn't sign up eight new agents right away. Based on the results, a more targeted approach was suggested: moving two agents from Lakeview to Greenfield, hiring two more at Hillcrest, and implementing the hold-reduction programme at Riverside.

There are twelve agents required in Greenfield, which has 88.64% utilisation. Lakeview can transfer 2 agents without exceeding its forecast capacity of 56.17% utilisation. Both hubs are in the same cluster, thus the transfer is allowed and will cost BDT 4,000.

Riverside has 90.80% utilisation and 64.42% of its holds are agent side. Before I would hire here, I would have to spend BDT 15 000 for reducing these holds. At Hillcrest, the utilisation rate is 95.87% and they require 11 agents for a total of 2 fixed agents, which comes to BDT 43,800 consisting of monthly salaries and hiring fees.

The forecasts are already adequate and much fuller than what they need for Brookside and Westwood. Instead of hiring more employees, Brookside should look into its three agents that have high returns. Planned expenses of the proposed plan are BDT 62,800 per month, based on the salaries quoted. It assumes that the programme brings improvement over a fourteen day period and that new team members have a target of 60% efficiency in their first week so that the target may be exceeded during this period.

Q9. The Budget

I would spend BDT 33,800 on two new agents' monthly salaries, BDT 10,000 on hiring, BDT 4,000 on transfers, and BDT 15,000 on Riverside's programme. This totals BDT 62,800, leaving BDT 7,200 as a reserve. Paying the new agents for all 28 forecast days would increase the total to BDT 65,400.

If the programme improves gradually as assumed, it could save approximately BDT 25,682 **in** hold costs over four weeks. These savings are estimates, not guaranteed results.

Q10. The Risks

Orders could exceed the forecast, new agents could learn slowly, or the hold programme might not work as expected. I would check workload, holds, returns, backlog, and spending daily, with reviews after seven and fourteen days. Riverside may remain slightly above the 85% target during the first two weeks.